

Township Infrastructure Management System (TIMS)

Complete Actor Roles, Functionalities, Modules & End-to-End Workflow

Project Scope

TIMS is a township-level infrastructure management platform that manages the complete lifecycle of infrastructure issues: reporting, validation, triage, contractor estimation, AMC-based approval, execution, billing, payment authorization, and resident verification.

The system uses Annual Maintenance Contracts (AMCs) with predefined contractor rates. TIMS therefore manages contractual rates, work quantities, approvals and billing rather than township-owned inventory.

1. System Actors

Actor	Type	Primary Responsibility
RWA Representative	External	Report infrastructure issues, track progress, and verify completed work.
Field Contractor	External	Inspect issues, prepare AMC-based estimates, execute repairs, and submit completion evidence.
Desk Clerk	Internal	Validate complaints, perform triage, create work orders, assign contractors, and monitor queues.
Department Head	Internal	Supervise department operations, handle escalations, and approve applicable budgets.
Finance Clerk	Internal	Verify estimates/invoices against AMC rates and authorize payment.
Township COO	Internal	Monitor township-wide KPIs, expenditure and approve major financial requests.
System Administrator	Internal	Manage users, roles, departments, township configuration, contractors, AMCs and system settings.

2. RWA Representative

The RWA Representative is the external representative connecting residents with township infrastructure operations.

Report an Issue

- Create a complaint with category, description, location, severity and photographs.
- Select the relevant infrastructure domain such as Civil, Electrical or Water.

Track Complaints

- View the current ticket status, assigned contractor, expected completion and progress.

View Resolution Evidence

- View before/after photographs and completion information submitted by the contractor.

Verify Completed Work

- Confirm that the issue has been resolved.
- Dispute or reopen the ticket if the work is incomplete or unsatisfactory.
- A defined auto-close period can be used when the RWA does not respond.

3. Field Contractor

The Field Contractor performs technical inspection and ground-level execution under the applicable AMC.

Receive Work Orders

- View assigned work orders and their priority/SLA information.

Inspect the Site

- Visit the location and determine the actual problem and required work.
- Record required labour, work quantities and applicable contractual items.

Prepare AMC-Based Estimate

- Select predefined contractual rate items from the applicable AMC.
- Enter quantities or labour hours; TIMS calculates the estimate using the contractual rate.
- The contractor does not arbitrarily enter a new rate for an AMC-covered item.

Execute Work

- Update the work through Assigned → In Progress → Completed.
- Record actual work/quantities and upload mandatory before/after evidence.

Handle Reopened Work

- Respond to RWA disputes and perform corrective work when a completed ticket is reopened.

4. Desk Clerk

Desk Clerks are operational users responsible for departmental triage and work-order coordination. Separate clerks may operate for Civil, Electrical and Water.

Complaint Validation

- Review incoming complaints, photographs, location and selected category.
- Check whether the complaint belongs to the correct department.

Duplicate Detection

- Use location-based duplicate detection to associate nearby reports with an existing master ticket when appropriate.

Work Order Creation

- Convert a validated complaint into a work order.
- Assign the work to an appropriate empanelled AMC contractor.

Queue Monitoring

- Monitor pending, assigned, in-progress, completed and SLA-at-risk work.

Escalation

- Escalate SLA breaches, technical exceptions, contractor delays and other issues to the Department Head.

5. Department Head

The Department Head manages one infrastructure department and acts as the main managerial approval/escalation layer.

Department Monitoring

- Monitor open work, SLA compliance, pending approvals, completion and expenditure.

Escalation Handling

- Resolve field exceptions, contractor issues and SLA breaches.

Budget Approval

- Approve estimates that fall within the department's configured financial authority.
- Authorize budget overrides where required.

Major Approval

- For requests above the configured major-expenditure threshold, participate in joint approval with the Township COO.

6. Finance Clerk

The Finance Clerk performs financial verification after work is completed.

Estimate and Invoice Verification

- Compare the approved estimate, actual work/quantity and invoice.

AMC Rate Verification

- Check that billed rates correspond to the applicable AMC/rate card.

Variance Handling

- Flag differences between expected and actual quantities or costs for investigation.

Payment Authorization

- Once financial checks pass, authorize the invoice for payment processing.

7. Township COO

The Township COO is the top operational management role and focuses on township-wide performance rather than individual routine tickets.

Township Dashboard

- View total complaints, open work, SLA compliance, resolution times, departmental performance and expenditure.

Major Financial Approval

- Approve major expenditure requests that require COO-level authorization.

Performance Monitoring

- Review department and contractor performance using aggregated operational data.

Strategic Oversight

- Monitor township-wide trends in infrastructure issues, spending and service performance.

8. System Administrator

The System Administrator manages the TIMS platform and its configuration rather than day-to-day infrastructure execution.

User and RBAC Management

- Create, disable and manage user accounts.
- Assign roles and permissions.

Township Configuration

- Manage township sectors, blocks, departments, categories and system-level configuration.

Contractor and AMC Administration

- Maintain contractor profiles, AMC validity, contract details and contractual rate cards.

Workflow Configuration

- Configure SLA rules, severity rules and approval thresholds.

Audit and Security

- Review important system actions, status changes, approvals and configuration changes.

9. Township & Location Hierarchy

Township → Sector/Zone → Block/Street → Infrastructure Asset → Complaint → Work Order

- The township is the top-level operational boundary.
- Sectors/zones and blocks provide more precise geographic organization.
- Infrastructure assets can be associated with a location and department.
- Complaints can be linked to the relevant asset/location.

10. Infrastructure / Asset Management

TIMS should maintain a basic record of township infrastructure so that complaints are connected to known assets rather than existing only as isolated tickets.

- Civil: roads, drains, footpaths, manholes and public structures.
- Electrical: street lights, transformers, electrical panels and related infrastructure.
- Water: pipelines, pumps, valves, tanks and leakage-related infrastructure.
- Asset information can include Asset ID, type, location, department, condition, installation information and maintenance history.

11. Annual Maintenance Contract (AMC) Management

AMC management is a central part of the financial model. Contractors provide predefined rates in their signed contract with the township.

- Contractor name and contract number.
- Department/service domain.
- Contract start and end dates.
- Rate card containing labour, service and work-item rates.
- Applicable terms and conditions.

- Contract status and renewal/expiry information.
- Historical AMC versions so older work orders continue referencing the contract/rates applicable at the time.

Example AMC Rate Calculation

Item	Quantity	Contract Rate	Amount
Electrician	2 hours	■500/hour	■1,000
General Labour	3 hours	■300/hour	■900
Cable Work	10 m	■150/m	■1,500
Total			■3,400

12. Complete End-to-End Workflow

1. Reporting

RWA Representative submits a complaint with category, description, location, severity and photographs.

2. Location & Duplicate Check

TIMS validates the location and checks whether a nearby existing issue should be associated with the new report.

3. Desk Triage

The relevant departmental Desk Clerk validates the complaint and determines whether a work order should be created.

4. Work Order Creation

A validated issue becomes a work order and is assigned to an appropriate AMC contractor.

5. Contractor Inspection

The contractor visits the site, inspects the issue and records the required work/quantities.

6. AMC-Based Estimation

The contractor selects applicable contractual rate items. TIMS calculates cost using AMC rates and entered quantities/hours.

7. Financial Approval

The system evaluates the estimate against configured approval thresholds and routes it to the appropriate authority.

8. Work Execution

After approval, the contractor moves the work to In Progress, performs the repair and records actual work/quantities.

9. Completion Evidence

The contractor uploads after-work evidence and marks the work Completed.

10. Finance Verification

Finance compares approved estimate, actual quantities and invoice against the applicable AMC rate card.

11. Payment Authorization

A financially valid invoice is authorized for payment.

12. RWA Verification

The RWA Representative confirms the repair or disputes/reopens it.

13. Closure

Confirmed work is closed. A defined no-response period may trigger automatic closure.

14. Rework Loop

A disputed ticket returns to the contractor for corrective action and repeats the completion/verification process.

13. Suggested Approval Flow

Estimated Amount	Approval Route
Within configured auto-approval threshold	System auto-approval, if permitted by policy.
Department-level threshold	Department Head approval.
Major expenditure threshold	Department Head + Township COO joint approval.

The exact monetary thresholds should be treated as configurable system parameters and finalized according to the township's requirements.

14. Work Order Status Lifecycle

REPORTED → UNDER_REVIEW → VALIDATED → WORK_ORDER_CREATED → ASSIGNED → INSPECTION → ESTIMATE_SUBMITTED → PENDING_APPROVAL → APPROVED → IN_PROGRESS → COMPLETED → PENDING_VERIFICATION → VERIFIED → CLOSED

Exception path: PENDING_VERIFICATION → DISPUTED → REOPENED → IN_PROGRESS → COMPLETED.

15. Core TIMS Modules

1. User & Role-Based Access Control (RBAC)
2. Township, Sector, Block and Location Management
3. Infrastructure / Asset Management
4. Complaint Management
5. Work Order Management
6. Contractor Management
7. Annual Maintenance Contract (AMC) Management
8. Approval Management
9. Finance & Billing
10. RWA Verification & Dispute Management
11. Notifications and SLA Alerts
12. Analytics & Dashboards
13. Audit Logs

16. Scope Boundary

Included: infrastructure issue lifecycle, AMC-based pricing, contractor work orders, approvals, financial verification, payment authorization, RWA verification, analytics, RBAC and auditability.

Not a core module: township-owned inventory/warehouse/resource-stock management. The contractor operates under the AMC and bills according to the applicable contractual rates and actual approved work.

17. Overall System Chain

Township → Sector → Block → Infrastructure Asset → Complaint → Work Order → AMC Estimate → Approval → Execution → Invoice → Finance Verification → RWA Verification → Closure